

**DEPARTMENT OF THE ARMY
JUSTIFICATION AND APPROVAL
FOR OTHER THAN FULL AND OPEN COMPETITION**

1. Contracting Activity: Army Contracting Command-Detroit Arsenal (ACC-DTA), Detroit Arsenal, Michigan is the procuring activity for this action.

2. Description of Action: ACC-DTA proposes to award a ten-year requirements contract (five one-year ordering periods and five one-year option ordering years) with Caterpillar Defense (Caterpillar) of Peoria, Illinois for Service Life Extension Program (SLEP) work on Caterpillar Earthmoving equipment. The planned contract also includes [REDACTED]

[REDACTED] and language to enable application of emerging Maintenance Work Orders (MWOs) and Engineering Change Proposals (ECP) that might impact the various vehicle configurations. The contract type will consist of Fixed Price with Economic Price Adjustment (FP with EPA) and Firm-Fixed Price (FFP) completion Contract Line Item Numbers (CLINS), to be funded with with Fiscal Year 2023-2033 (FY23-33) Other Procurement, Army (OPA) funds.

3. Description of Supplies/Services: The proposed action will be executed on behalf of the Product Manager (PdM), Engineer, Maintenance and Support Equipment (EMSE). The candidate vehicle types for SLEP are: Vibratory (Vibe) Rollers I, II, and III; High Speed Compactors (HSC); Dual Steel Wheel Rollers (DSWR); D7G, D7R, and D6K Dozers; 130G and 120M Graders; M966H, 924G, and 924H Loaders; 613C, 621B and 621G Scrapers; and the 613C Water Distributor. The SLEP process results in a like new overhaul of candidate vehicles that are inducted, and includes mandatory parts replacement, excess property remanufacture or disposal, and Chemical Agent Resistant Coating (CARC) paint application. The contractor will be required to perform the SLEP effort on an estimated quantity of 256 Government furnished vehicles during the ten-year contract period, expected to commence by the end of September 2023 and conclude by September 2033. Approximately 123 of those vehicles may receive configuration updates as part of the SLEP. The SLEP effort consists of two parts: 1) A Core Effort (Core) which includes only those tasks that are performed on each vehicle, and 2) Additional Work Effort (AWE), which includes repair actions beyond the Core Effort that are unique to each individual vehicle based on its condition upon initial inspection conducted by the contractor. Once the vehicle's condition is assessed, the initially funded average AWE threshold is adjusted to the actual amount required per vehicle. The purpose of SLEP effort is to return the vehicles to a near-zero hours/zero miles condition. The SLEP effort is projected to double the estimated useful life of the vehicles by an additional 10 to

15 years. Vehicles will be identified based on mission needs and availability, with the PdM EMSE providing the list of vehicles to be inducted to the SLEP program with a pickup/drop off point of contact. Based on the Independent Government Cost Estimate (IGCE), prepared by PdM EMSE on 18 JAN 2023, the estimated dollar value of this contract is [REDACTED]

4. Authority Cited: The statutory authority permitting other than full and open competition is 10 United States Code (U.S.C.) 3204 (a)(1), as implemented by Federal Acquisition Regulation (FAR) 6.302-1, "Only One Responsible Source and No Other Supplies or Services Will Satisfy Agency Requirements." The specific citation applied is FAR 6.302-1(a)(2)(ii) "Supplies may be deemed to be available only from the original source in the case of a follow-on contract for the continued development or production of a major system or highly specialized equipment, including major components thereof, when it is likely that award to any other source would result in (A) a substantial duplication of cost to the Government that is not expected to be recovered through competition."

5. Reason for Authority Cited: The aforementioned citations are applicable because the proposed action is considered a follow-on effort to the previous SLEP contract W56HZV-20-D-0094 awarded to Caterpillar. The candidate vehicle types were produced by Caterpillar under prior ACC-DTA contracts, and some have undergone previous overhaul efforts as well. [REDACTED]

[REDACTED] The Government does not have the pertinent technical data to enable a competitive re-procurement. As the Original Equipment Manufacturer (OEM), Caterpillar retains the technical data rights to the vehicles. [REDACTED]

[REDACTED] Caterpillar will not sell its vehicle technical data to the Government because it contains proprietary data. On 31 JAN 2023, ACC-DTA inquired if Caterpillar was willing to sell the Technical Data Package (TDP) for the vehicles listed in Paragraph 2 above. On 2 FEB 2023, Caterpillar responded, stating the following: "Caterpillar is not interested in selling Technical Data Packages for our product line as we deem this Proprietary Intellectual Property for our commercial business."

a. As Caterpillar is unwilling to sell the TDP rights to the Government, any other source would be required to reverse engineer and test a new vehicle design, or to source substantially all of the parts necessary for SLEP from Caterpillar, which would result in a substantial duplication of cost to the Government that is not expected to be recovered through competition. This determination is based on the following facts:

Unique Qualifications. The SLEP process is very complex as it requires the vehicle to be broken down to its most basic level, and requires the contractor to perform reliability analysis and replacement recommendations based on its

engineering and logistical knowledge. Caterpillar is in the unique position to perform this work because of its direct manufacturing, extensive dealer network and supply chain, field support, and historical experience performing SLEP on these vehicles. Caterpillar has extensive knowledge from past experience that it leverages during performance. This knowledge includes: design, manufacturing engineering, test, and user data; service bulletins; and supply chain data. Therefore, any other source would have a significant learning curve before the risk of unsuccessful performance would be at an acceptable level. Caterpillar has successfully completed SLEP efforts on all required vehicle types, under multiple delivery orders issued against the previous contracts.

Duplication of Cost: Any source other than Caterpillar would need to reverse engineer the vehicles, create the needed design and manufacturing data, and fully test the item in order to perform this effort. No other source possesses the original assembly specifications, interface requirements, or tolerance requirements to which these vehicles were manufactured. An engineering estimate developed by the U.S. Army Combat Capabilities Development Command (CCDC), Ground Vehicle Systems Center (GVSC), indicates it would cost [REDACTED] to reverse engineer one vehicle type. The scope of the calculated estimates include an average quantity of [REDACTED] parts per vehicle, data development, production planning and tooling, testing efforts, and TDP maintenance. The table below provides a cost estimate breakdown:

Table II: Duplication of Cost

Reverse Eng. Cost Estimate (Avg. 6,000 parts per vehicle platform, total of 8 vehicle platforms with multiple variants)

i - Remove Parts/Disassemble Machine	[REDACTED]
ii - Technical Data Package (TDP)/Part Drawings for Fabrication	[REDACTED]
iii - Production Planning and Tooling	[REDACTED]
iv - Testing (Validation & Verification)	[REDACTED]
v - TDP Maintenance	[REDACTED]
Total Estimated Cost to Reverse Engineer Vehicle	[REDACTED]

- i. Removal of Parts/Disassemble: To reverse engineer any of the Earthmoving vehicle types, any contractor, other than the OEM would need to fully disassemble the machine to catalog the parts, identify the specifications, and take measurements of the parts and tolerances of the parts in relation to their location on the equipment. Removing parts and disassembling the machine is a labor intensive effort, PdM EMSE Engineering estimates it would require two Heavy Equipment Mechanics 2,020 hours each to fully complete the removal/disassembly process, based on known efforts with similar types of equipment. This effort is estimated to cost [REDACTED] per vehicle platform, for a total cost of [REDACTED] for the eight vehicle platforms planned to undergo SLEP.

- ii. TDP Development: A contractor other than the OEM would have to develop part drawings for fabrication of the replacement parts. These drawings would be utilized to build the tooling necessary to produce the parts for the equipment. Based on similar efforts, it is estimated the cost for [REDACTED] for the eight vehicle platforms planned to undergo SLEP.
- iii. Production Planning and Tooling: Production Planning and Tooling are the costs associated with the production of the parts necessary to complete the SLEP effort by a contractor other than the OEM. To complete this effort a contractor, other than the OEM, would need to establish a labor force trained and capable of producing the parts needed. In addition, the contractor would potentially need to acquire: a production facility, tooling, support equipment, and the machinery needed to maintain the equipment. Based on similar commercial equipment efforts, the cost of production planning and tooling is estimated to be [REDACTED] per vehicle platform, for a total cost of [REDACTED] for the eight vehicle platforms planned to undergo SLEP.
- iv. Testing: Vehicles undergoing the SLEP process by a contractor other than Caterpillar will have to undergo contractor testing (validation) and Government testing (verification). This is required to assure the product conforms to the Purchase Description (PD). The estimate for validation is [REDACTED], while the estimate for verification costs is [REDACTED], which are both based on similar efforts. The Government's cost to perform verification efforts is significantly lower as the validation is performed first and generally eliminates the need for the Government to perform duplicative testing efforts. In addition to the testing efforts, the contractor must provide subject matter experts to support Government testing. The estimated cost for contractor test support is [REDACTED] over and above the estimated costs of validation and verification testing. The required re-testing would result in duplicated costs of [REDACTED] for validation, [REDACTED] for verification, and [REDACTED] for contractor test support for a total duplicated cost of [REDACTED] for testing per vehicle platform, for a total cost of [REDACTED] for the eight vehicle platforms planned to undergo SLEP.
- v. TDP Maintenance: Once established, the TDP will require maintenance to ensure it is current and accurate to the current version of the Earthmoving vehicles. Because Caterpillar sells commercial versions of these machines in the commercial market, any changes to configuration are updated and incorporated at the time of change. Those costs are then absorbed by both the Government and commercial business with Caterpillar. A contractor other than Caterpillar would have to update the TDP after the change is made and ensure the change conforms to the PD. Those costs would be a direct charge to the Government. Based on similar actions, TDP maintenance is estimated to cost [REDACTED] per vehicle platform, for a total cost of [REDACTED]

█████ for the eight vehicle platforms planned to undergo SLEP .

- vi. **Total Estimated Duplicated Cost:** The total duplicated costs of █████ across eight different platforms, not even considering the variety of models of each, exceeds the expected █████ cost for this SLEP effort by 230%. If the Government were to compete this requirement, it is not possible that the Government could achieve savings through competition sufficient to offset the estimated duplication of costs necessary for a source other than Caterpillar to be able to perform the SLEP effort. As such, if the Government were to compete this effort and award to a source other than Caterpillar, it is certain there would be a substantial duplication of cost, which is not expected to be recovered through subsequent competition.

6. Efforts to Obtain Competition:

- a. **Effective Competition:** Competition will not be pursued for this action for reasons stated in section 5. Pursuant to FAR 5.201, this proposed contract action will be synopsized on www.SAM.gov and all bids or proposals received will be considered by the Government.
- b. **Subcontracting competition:** Caterpillar is considered a large business and will be required to submit a Small Business Subcontracting Plan for this contract. In addition, the contract includes FAR clauses 52.244-5 "Competition in Subcontracting," and 52.244-2 "Subcontracts", to ensure Caterpillar competitively awards subcontracts to the maximum extent possible.

7. Actions to Increase Competition: The Government does not plan to seek additional competition for this effort. As detailed in section 5 above, the costs to create a competitive SLEP program would result in a substantial duplication of cost which the Government does not expect to recover through a competitive procurement. The Government will continue to survey the marketplace and will review its acquisition strategy should additional requirements or additional sources be identified.

8. Market Research: On 19 May 2022, a Sources Sought Notice (SSN) was posted for a period of 30 days to beta.SAM.gov by ACC-DTA on behalf of PdM EMSE in accordance with FAR 10.002. The purpose of the questionnaire was to solicit responses from industry that would allow the Government to assess whether or not capabilities exist that could accomplish the proposed acquisition. The research included questions regarding the ability of contractors to acquire proprietary parts, and whether any contractors have detailed knowledge of the original assembly specifications, interface requirements, and tolerances required to meet Government requirements. Two parties responded to the sources sought, the OEM, Caterpillar and one other potential contractor, Atlantic Diving Supply (ADS) of Virginia, partnered with one of the Caterpillar SLEP subcontractors,

Carter Machinery, Inc. and proposing to leverage Red River Army Depot (RRAD).

Caterpillar indicated a broad dealer network across the continental United States which can perform SLEP to like new specification. The contractor stated it has almost limitless capacity within a 400-mile radius to any end-user base/facility to efficiently rebuild machines and components with its ability to engage its dealer network. Caterpillar stated that it has performed SLEP work on 8-15 year old machines for U.S. Army, U.S. Navy and U.S. Marine Corp. under multiple contracts dating back to 2010.

ADS proposed a team approach with itself providing "total contract management support," RRAD providing "disassembly and assembly of the vehicles" and Carter Machinery providing "all the parts, component rebuilds, and technical support." ADS indicated it currently manages a "Caterpillar Commercial Price List contract, providing program management and parts support." Carter Machinery has recent experience performing SLEP work as a subcontractor to Caterpillar, and RRAD has performed maintenance on similar vehicles though not to near-zero mile specifications. ADS as a prime contractor for SLEP would be required to complete a pilot effort with its partners, Carter Machinery and RRAD, at a substantial additional cost and schedule impact to the program, before proceeding to overhaul Earthmoving vehicles in larger quantities.

In order to perform a pilot or larger quantity SLEP, The ADS team would be dependent on the vehicle OEM for their supply chain in order to obtain the mandatory and condition based replacement components, and therefore unable to offer any cost advantage on materials over the current contract with Caterpillar. In addition, the ADS proposal to enlist RRAD does not mitigate these competitive disadvantages. An effort employing RRAD would also be subject to a lengthy pilot effort, in the absence of the proprietary vehicle specifications, and with increased cost risk due to the labor rate structure of the Army organic industrial base, which fluctuates annually based on workload. RRAD is principally a source for RESET of vehicles to 10/20 standards, which is vastly different from restoring a vehicle to near zero miles and original factory specifications. Caterpillar may engage Carter Machinery as a subcontractor for the Earthmoving SLEP, depending on the proximity of Carter Machinery locations to the pick-up location of the seed assets planned to undergo SLEP, which is preferred kept under a 400 mile radius to minimize transportation costs to the Government.

9. Interested Sources: Other than the OEM and ADS who responded to the SSN, no other contractor has expressed interest in performing this effort.

10. Other Facts:

a. *Procurement History:* ACC-DTA previously acquired pilot and production level SLEP for many of the candidate vehicle types. The previous contracts are described below:

- (1) Contract W56HZV-15-D-0003 was awarded in November 2014 for the SLEP of the Deployable Universal Combat Earthmover (DEUCE). Modification P00003 to contract W56HZV-15-D-0003, awarded 23 September 2016, established four ordering years for Compaction vehicle SLEP requirements. Contract W56HZV-20-D-0094 was awarded 30 September 2020 for additional Compaction vehicle SLEP, and was later modified to also include SLEP of the D7G Dozer.
- (2) All of the above contract actions were awarded on a sole source basis.
- (3) The authority cited in each case was FAR 6.302-1.
- (4) The most recent Justification and Approval document, which was for the addition of the D7G Dozer to the Compaction vehicle SLEP contract, W56HZV-20-D-0094, indicated under paragraph 7 that the Government did not plan to seek additional competition, but would continue to survey the marketplace for opportunities to do so.
- (5) None of the previous contract SLEP actions were based on full and open competition.
- (6) There are no apparent unusual patterns with the previous contract actions.
- (7) The circumstances supporting sole source procurement of SLEP efforts have remained the same for several years. Caterpillar, as the OEM for the subject vehicle types, retains the technical data, expertise, supply chain and dealer network necessary to perform the work.

b. Other Facts: None.

11/12. Technical and Requirements Certification:

I certify that the supporting data under my cognizance which are included in the justification are accurate and complete to the best of my knowledge and belief.

Signature: //KMB//	13 March 2023
Kyle Bruner Project Manager, Force Projection	Date:

13. Fair and Reasonable Cost Determination:

I hereby determine that the anticipated cost to the Government for this contract action will be fair and reasonable. The contractor's proposal will be reviewed by Government requirements personnel who will address the appropriateness of the proposed labor hours and labor categories, materiel, other direct costs, and other cost elements as requested by the Contracting Officer. Cost/Price analysis will be used to discern whether the costs proposed are current and justifiable. The final negotiated agreement will give full consideration to all contractual requirements, including the delivery

schedule, to support a determination of fair and reasonable prices for this action. The contractor will be required to submit Certified Cost and Pricing Data, which will form the basis for negotiating a fair and reasonable price.

Signature: //s, LG//	
Lisa Gerasimidis Procuring Contracting Officer	Date: 24 March 2023

14. Contracting Officer Certification:

I certify the facts and representations under my cognizance which are included in the J&A are accurate and complete to the best of my knowledge and belief.

Signature: //s, LG//	
Lisa Gerasimidis Procuring Contracting Officer	Date: 24 March 2023

APPROVAL

Based on the foregoing justification, and as Senior Contracting Official pursuant to Army Materiel Command Head of the Contracting Activity Delegation of Authority #21-211 hereby approve the procurement of Caterpillar Service Life Extension Program (SLEP) for the Earthmoving vehicle fleet on an other than full and open competition basis pursuant to the authority of 10 U.S.C.3204(a)(1), as implemented by FAR 6.302-1; "Only One Responsible Source and No Other Supplies and Services Will Satisfy Agency Requirements". This approval is subject to availability of funds, and provided that the services and property herein described have otherwise been authorized for acquisition. The total estimated value of this action is \$87M.

4/18/2023

X Daniel J. Gallagher

Daniel J. Gallagher

Senior Contracting Official

Signed by: GALLAGHER,DANIELJOSEPH.1034297360